

Question	Marking Guidance	Mark	Comments
05.1	1. Wash hands with soap OR Disinfect surfaces; 2. Use sterile pipette/syringe (to transfer bacteria); 3. (Remove bottle lid and) flame neck of bottle; 4. Lift lid of (agar) plate at an angle; 5. Work close to upward air movement; 6. Use sterile spreader; 7. Place pipette/spreader into disinfectant (immediately after use);	3 max (3 x AO1)	1. Ignore sterilise hands OR surfaces 1. Accept sanitise for disinfect 1. Accept antiseptic /antimicrobial/alcohol (wipes) 1. and 7 Accept a named type of disinfectant 2. Reject loop 2. Accept use unopened pipette/ syringe for sterile 4. Accept lift lid slightly OR keep lid over plate 4. Ignore 'work quickly with lid off' 5 Reject 'air movements sterilise air' 5 Accept 'convection current' for 'upward air movement' 6. Accept loop for spreader 6. Examples of sterilising technique eg, flame OR 'dip in alcohol and flame' OR 'dip in disinfectant and rinse (in sterile water)'

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05.2	Correct answer in range of 1.768 to 1.8 OR 2 = 2 marks Accept for 1 mark, evidence of 28/28.26/28.3 (correct calculation, $\pi \times 3^2$) OR 28.27433 (correct calculation using π value on calculator) OR 0.4423 (correct calculation using diameter rather than radius) OR 1.76 (correct calculation but incorrectly rounded);	2 (2 x AO3)	A common correct answer for 2 marks is 1.77 Accept correct rounding of figures

Question	Marking Guidance	Mark	Comments
05.3	(Positive control) 1. Antimicrobial/ antibacterial (solution) OR Antibiotic OR Antiseptic/disinfectant; (Negative control) 2. (Sterile) water OR Oil (without cinnamon);	2 (2 x AO3)	If not specified, accept the first answer as being for the positive control 1. Accept named antimicrobial/antibacterial/antibiotic/antiseptic/disinfectant

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05.4	<table><tr><td>Median for all cultures</td><td>16</td><td>12</td></tr><tr><td>Mean for all cultures</td><td>17</td><td>13</td></tr></table>	Median for all cultures	16	12	Mean for all cultures	17	13	1 (AO2)	Accept 16.6 and 13.4 as the mean values (in this order)
	Median for all cultures	16	12						
	Mean for all cultures	17	13						
All four numbers correct;									

Question	Marking Guidance	Mark	Comments
05.5	1. (Mean $\pm$ 2SD) 12.2 to 21.8 and 8.6 to 17.4 OR (Mean $\pm$ 2SD) 11.8 to 21.4 and 9(.0) to 17.8 OR (Mean $\pm$ 1.96 SD) 12.3 to 21.7 and 8.7 to 17.3 OR (Mean $\pm$ 1.96 SD) 11.9 to 21.3 and 9.1 to 17.7; 2. (SD) overlap so difference (likely to be) due to chance OR (SD) overlap so (likely) no significant difference (in means);	2 (2 x AO3)	1. Accept ECF for 1 mark, correct SDs calculated using incorrect means in 05.4 2. Accept ECF for 1 mark, correct explanation based on correct SDs calculated from incorrect means in 05.4 2. Reject results are due to chance OR results are significant
